Supplementary Appendix: Apparent Catamenial Epilepsy Classification Under Independence

This appendix provides additional methodological details, defined analysis choices, quality-control checks, sensitivity analyses, and supplementary results for the null simulation. It is intended as online supplementary material accompanying the main Neurology submission draft.

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A1. Simulation design and reproducibility

The analysis used a deterministic master seed of 20260505. The full run simulated 100,000 synthetic participants, processed in chunks, with 50,000 in the healthy ovulatory cohort and 50,000 in the heterogeneous menstruating-age cohort. The 36-month diaries were generated independently for seizures and menstrual/hormone cycles, followed by a participant-specific random circular shift before merging.

Daily audit data were saved for a 1% random sample. Participant-level and window-level outputs were written as parquet and CSV files, and every output artifact was recorded in a machine-readable manifest with file size and SHA-256 checksum.

The term 'defined before the full simulation run' is used instead of 'prespecified' unless a dated protocol, locked commit, or preregistration is supplied. The manifest records the seed, config values, output checksums, and adapter assumptions.

A2. Simulator descriptions and validation targets

CHOCOLATES generated seizure-count diaries independently of menstrual-cycle state. The adapter records observed seizure burden and the simulator-reported dominant seizure-cycle period when available. Because multidien rhythms near monthly periods are a key potential mechanism for chance alignment, submitted versions should report the distribution of dominant_seizure_cycle_days and sensitivity analyses excluding or flattening 24-35-day seizure rhythms.

HORMONE-CYCLE generated menstrual/hormone diaries independently of seizure state. The adapter records age, cycle length, within-participant cycle-length SD, ovulatory fraction, progesterone and estradiol when available, and inadequate-luteal-phase flags based on a progesterone threshold of 5.0 ng/mL. Submitted versions should include package versions or commits for both simulators and a validation table comparing simulated diary metrics with empirical benchmarks.

Supplementary Table 1. Minimum simulator validation targets for submission.

Domain	Simulated metric to report	Reviewer-facing purpose
Seizure burden	Seizure days/month, seizures/month, interseizure interval dispersion, seizure-cluster metrics	Shows the null seizure diary is not an unrealistically sparse or overclustered artifact.
Seizure rhythms	Dominant multidien period distribution; proportion in 24-35-day band; sensitivity excluding that band	Separates finite-count ratio artifacts from monthly-ish seizure-rhythm alignment.
Menstrual cycles	Cycle-length distribution, within-participant cycle-length SD, luteal length, ovulatory fraction, anovulation frequency	Shows phase labeling and C3 eligibility are generated from plausible cycle structure.
Missing/irregular behavior	Unlabeled days, non-23-35-day cycles, ILP/anovulatory flags, medical-factor prevalence	Quantifies sources of indeterminate windows and C3 sensitivity.

A3. Cohort definitions and diary generation

The healthy ovulatory cohort targeted adult ovulatory cycling, age 18-45 years, with simulator medical modifiers disabled where exposed. The heterogeneous menstruating-age cohort targeted a broader menstruating-age range of 13.0-54.9 years and allowed anovulation and irregularity. Because the hormone simulator exposed medical-factor controls but not a natural prevalence sampler, PCOS, peri-menarche, perimenopause, and dysmenorrhea were sampled from config.yaml rates. This cohort is an assumption-driven heterogeneity stress test unless external prevalence calibration is added.

Supplementary Table 2. Cohort construction parameters.

Parameter	Healthy ovulatory	Heterogeneous menstruating-age
Target sample size	50,000	50,000
Age range	18.0-45.0 years	13.0-54.9 years
Ovulatory-cycle handling	Ovulation probability set to 1.0	Natural ovulatory/anovulatory behavior allowed
PCOS rate	Disabled	10%

Parameter	Healthy ovulatory	Heterogeneous menstruating-age
Peri-menarche rate	Disabled	60% when age <20 years
Perimenopause rate	Disabled	55% when age >=45 years
Dysmenorrhea rate	Disabled	12%

A4. Phase labeling and catamenial epilepsy definitions

Herzog-style phases were assigned on the full diary before window subsetting. For a complete cycle of length L, day 1 was the onset of menstrual flow and backward day was defined as d-(L+1). Strict Herzog phase assignment used an ordered lookup: menstrual phase if days 1-3 or backward days -3 to -1; follicular phase if days 4-9; ovulatory phase if day >=10 and backward day <=-13; luteal phase if backward days -12 to -4. The ovulatory condition is the intersection of forward and backward criteria. Days not satisfying any rule were unlabeled. The luteal-anchored sensitivity preserved the same menstrual and luteal windows but fixed ovulatory phase to backward days -16 to -13 and labeled remaining nonmenstrual, nonovulatory, nonluteal days as follicular.

Supplementary Table 3. Implemented catamenial epilepsy definitions.

Definition	Primary rule	Main analyzability condition
Exact Herzog 2004	Three-cycle legacy rule; any CE if at least 2 of 3 cycles show any C1, C2, or C3 pattern.	Only complete 3-cycle windows with strict 23-35 day cycles.
Windowed Herzog thresholds	C1 ADSF(M)/ADSF(F+L) >=1.69; C2 ADSF(O)/ADSF(F+L) >=1.83; C3 ADSF(O+L+M)/ADSF(F) >=1.62 among ILP-flagged days.	Required phases observed; undefined ratios marked indeterminate.
Minimum-data rule	Windowed Herzog thresholds after applying minimum duration and seizure-burden criteria.	At least 4 months or 6 complete cycles, plus at least 4 seizure days.

Definition	Primary rule	Main analyzability condition
Cycle reproducibility	At least two-thirds of eligible cycles positive for the same pattern and pooled ratio passes threshold.	At least 6 complete cycles for primary rule; 12 cycles for sensitivity.
Negative-binomial regression	Daily GLM with M and O indicators; one-sided Holm-adjusted tests and C1/C2 rate-ratio thresholds. Stabilized and window-only dispersion versions reported.	Minimum-data rule passed; Poisson robust fallback recorded when needed.
Historical rules	Assumption-based operationalizations of Newmark-Penry, Duncan 1993, Herzog 1997 twofold, and Reddy 2007 any-phase rules.	Reported separately and flagged as assumption based.

C3 was evaluated only in the heterogeneous cohort when ilp_flag was true. Anovulatory cycles without ILP labeling were not automatically considered C3-positive or C3-applicable. Supplementary sensitivity rows report any CE excluding C3 and C1, C2, and C3 separately.

A5. Observation windows and analyzability rules

Calendar windows used one random valid start day per participant for 1, 3, 4, 6, 9, 12, 18, 24, and 36 months. Cycle windows used one random valid start among complete cycles for 3, 6, and 12 cycles. The full-window analysis used the entire 36-month diary. Windows were marked indeterminate, not negative, when required phases or denominator information were unavailable. Exact Herzog 2004 is rule-defined only for complete 3-cycle windows with strict 23-35-day cycles. Other window settings are not evaluated for this rule; they should not be interpreted as negative evidence.

A6. Statistical analysis and edge-case handling

All ratio definitions used average daily seizure frequency rather than raw counts. If a comparator phase had observed days but zero seizures, a positive numerator produced an infinite ratio and counted positive; zero numerator and zero comparator produced an undefined ratio and an indeterminate result. Wilson 95% confidence intervals were used for simple proportions. Study-level Monte Carlo sampled 10,000 studies of 30, 50, and 100 participants per cohort for the 3-month window.

Negative-binomial regression used daily rows with an intercept, menstrual indicator, ovulatory indicator, and cycle fixed effects when at least four complete cycles were available. The offset was one observed day. The Holm family comprised the menstrual and ovulatory one-sided tests within participant. Positivity required adjusted $P<0.05$ and the corresponding threshold ratio. NB regression was evaluated for full-diary windows in this revision. The stabilized-dispersion analysis uses full-diary alpha; the full-window, window-only alpha analysis is a non-oracle sensitivity.

Monte Carlo binomial intervals quantify uncertainty under the fixed simulation model. They do not include uncertainty about simulator validity, medical-factor prevalence, seizure-rhythm periods, or menstrual-cycle parameterization. Participant-level summary rows, all-attempted rates, indeterminate reasons, and unstable-denominator flags were generated from window_results.parquet. Rows with fewer than 1,000 classifiable windows are flagged as unstable and not interpreted.

A7. Supplementary results

Supplementary Table 3. Window-length sensitivity for core definitions.

Cohort	Observation window	Definition	Classifiable windows	False-positive windows	False-positive rate	Indeterminate
healthy ovulatory	1 month	Cycle reproducibility >=2/3, C1/C2	0	0	NA	100.0%
healthy ovulatory	1 month	Exact Herzog 2004, any pattern	0	0	NA	100.0%
healthy ovulatory	1 month	Herzog C1/C2 plus >=4 seizure-day minimum	0	0	NA	100.0%
healthy ovulatory	1 month	Negative-binomial regression C1/C2	0	0	NA	100.0%
healthy ovulatory	1 month	Windowed Herzog C1/C2 union	38,234	18,249	47.7%	23.5%
healthy ovulatory	1 month	Windowed Herzog thresholds	38,234	18,249	47.7%	23.5%
healthy ovulatory	3 months	Cycle reproducibility >=2/3, C1/C2	0	0	NA	100.0%
healthy ovulatory	3 months	Exact Herzog 2004, any pattern	0	0	NA	100.0%
healthy ovulatory	3 months	Herzog C1/C2 plus >=4 seizure-day minimum	0	0	NA	100.0%
healthy ovulatory	3 months	Negative-binomial regression C1/C2	0	0	NA	100.0%
healthy ovulatory	3 months	Windowed Herzog C1/C2 union	45,280	18,696	41.3%	9.4%
healthy ovulatory	3 months	Windowed Herzog thresholds	45,280	18,696	41.3%	9.4%
healthy ovulatory	4 months	Cycle reproducibility >=2/3, C1/C2	0	0	NA	100.0%
healthy ovulatory	4 months	Exact Herzog 2004, any pattern	0	0	NA	100.0%
healthy ovulatory	4 months	Herzog C1/C2 plus >=4 seizure-day minimum	37,641	13,501	35.9%	24.7%
healthy ovulatory	4 months	Negative-binomial regression C1/C2	0	0	NA	100.0%

Cohort	Observation window	Definition	Classifiable windows	False-positive windows	False-positive rate	Indeterminate
healthy ovulatory	4 months	Windowed Herzog C1/C2 union	46,381	17,894	38.6%	7.2%
healthy ovulatory	4 months	Windowed Herzog thresholds	46,381	17,894	38.6%	7.2%
healthy ovulatory	6 months	Cycle reproducibility >=2/3, C1/C2	6,706	704	10.5%	86.6%
healthy ovulatory	6 months	Exact Herzog 2004, any pattern	0	0	NA	100.0%
healthy ovulatory	6 months	Herzog C1/C2 plus >=4 seizure-day minimum	41,136	12,985	31.6%	17.7%
healthy ovulatory	6 months	Negative-binomial regression C1/C2	0	0	NA	100.0%
healthy ovulatory	6 months	Windowed Herzog C1/C2 union	47,489	16,185	34.1%	5.0%
healthy ovulatory	6 months	Windowed Herzog thresholds	47,489	16,185	34.1%	5.0%
healthy ovulatory	9 months	Cycle reproducibility >=2/3, C1/C2	19,087	666	3.5%	61.8%
healthy ovulatory	9 months	Exact Herzog 2004, any pattern	0	0	NA	100.0%
healthy ovulatory	9 months	Herzog C1/C2 plus >=4 seizure-day minimum	43,868	11,929	27.2%	12.3%
healthy ovulatory	9 months	Negative-binomial regression C1/C2	0	0	NA	100.0%
healthy ovulatory	9 months	Windowed Herzog C1/C2 union	48,361	14,169	29.3%	3.3%
healthy ovulatory	9 months	Windowed Herzog thresholds	48,361	14,169	29.3%	3.3%
healthy ovulatory	12 months	Cycle reproducibility >=2/3, C1/C2	16,143	255	1.6%	67.7%
healthy ovulatory	12 months	Exact Herzog 2004, any pattern	0	0	NA	100.0%

Cohort	Observation window	Definition	Classifiable windows	False-positive windows	False-positive rate	Indeterminate
healthy ovulatory	12 months	Herzog C1/C2 plus >=4 seizure-day minimum	45,228	10,677	23.6%	9.5%
healthy ovulatory	12 months	Negative-binomial regression C1/C2	0	0	NA	100.0%
healthy ovulatory	12 months	Windowed Herzog C1/C2 union	48,754	12,482	25.6%	2.5%
healthy ovulatory	12 months	Windowed Herzog thresholds	48,754	12,482	25.6%	2.5%
healthy ovulatory	18 months	Cycle reproducibility >=2/3, C1/C2	12,120	42	0.3%	75.8%
healthy ovulatory	18 months	Exact Herzog 2004, any pattern	0	0	NA	100.0%
healthy ovulatory	18 months	Herzog C1/C2 plus >=4 seizure-day minimum	46,767	8,587	18.4%	6.5%
healthy ovulatory	18 months	Negative-binomial regression C1/C2	0	0	NA	100.0%
healthy ovulatory	18 months	Windowed Herzog C1/C2 union	49,167	9,847	20.0%	1.7%
healthy ovulatory	18 months	Windowed Herzog thresholds	49,167	9,847	20.0%	1.7%
healthy ovulatory	24 months	Cycle reproducibility >=2/3, C1/C2	9,485	6	0.1%	81.0%
healthy ovulatory	24 months	Exact Herzog 2004, any pattern	0	0	NA	100.0%
healthy ovulatory	24 months	Herzog C1/C2 plus >=4 seizure-day minimum	47,550	6,965	14.6%	4.9%
healthy ovulatory	24 months	Negative-binomial regression C1/C2	0	0	NA	100.0%
healthy ovulatory	24 months	Windowed Herzog C1/C2 union	49,379	7,884	16.0%	1.2%
healthy ovulatory	24 months	Windowed Herzog thresholds	49,379	7,884	16.0%	1.2%

Cohort	Observation window	Definition	Classifiable windows	False-positive windows	False-positive rate	Indeterminate
healthy ovulatory	36 months	Cycle reproducibility >=2/3, C1/C2	6,132	1	0.0%	87.7%
healthy ovulatory	36 months	Exact Herzog 2004, any pattern	0	0	NA	100.0%
healthy ovulatory	36 months	Herzog C1/C2 plus >=4 seizure-day minimum	48,359	4,940	10.2%	3.3%
healthy ovulatory	36 months	Negative-binomial regression C1/C2	0	0	NA	100.0%
healthy ovulatory	36 months	Windowed Herzog C1/C2 union	49,605	5,576	11.2%	0.8%
healthy ovulatory	36 months	Windowed Herzog thresholds	49,605	5,576	11.2%	0.8%
healthy ovulatory	3 cycles	Cycle reproducibility >=2/3, C1/C2	0	0	NA	100.0%
healthy ovulatory	3 cycles	Exact Herzog 2004, any pattern	22,996	11,422	49.7%	54.0%
healthy ovulatory	3 cycles	Herzog C1/C2 plus >=4 seizure-day minimum	19	Suppressed	Not interpreted (<1,000)	100.0%
healthy ovulatory	3 cycles	Negative-binomial regression C1/C2	0	0	NA	100.0%
healthy ovulatory	3 cycles	Windowed Herzog C1/C2 union	45,048	18,481	41.0%	9.9%
healthy ovulatory	3 cycles	Windowed Herzog thresholds	45,048	18,481	41.0%	9.9%
healthy ovulatory	6 cycles	Cycle reproducibility >=2/3, C1/C2	22,669	2,779	12.3%	54.7%
healthy ovulatory	6 cycles	Exact Herzog 2004, any pattern	0	0	NA	100.0%
healthy ovulatory	6 cycles	Herzog C1/C2 plus >=4 seizure-day minimum	40,861	12,839	31.4%	18.3%
healthy ovulatory	6 cycles	Negative-binomial regression C1/C2	0	0	NA	100.0%

Cohort	Observation window	Definition	Classifiable windows	False-positive windows	False-positive rate	Indeterminate
healthy ovulatory	6 cycles	Windowed Herzog C1/C2 union	47,474	16,217	34.2%	5.1%
healthy ovulatory	6 cycles	Windowed Herzog thresholds	47,474	16,217	34.2%	5.1%
healthy ovulatory	12 cycles	Cycle reproducibility >=2/3, C1/C2	15,728	350	2.2%	68.5%
healthy ovulatory	12 cycles	Exact Herzog 2004, any pattern	0	0	NA	100.0%
healthy ovulatory	12 cycles	Herzog C1/C2 plus >=4 seizure-day minimum	45,112	10,618	23.5%	9.8%
healthy ovulatory	12 cycles	Negative-binomial regression C1/C2	0	0	NA	100.0%
healthy ovulatory	12 cycles	Windowed Herzog C1/C2 union	48,690	12,446	25.6%	2.6%
healthy ovulatory	12 cycles	Windowed Herzog thresholds	48,690	12,446	25.6%	2.6%
healthy ovulatory	36-month full diary	Cycle reproducibility >=2/3, C1/C2	6,132	1	0.0%	87.7%
healthy ovulatory	36-month full diary	Exact Herzog 2004, any pattern	0	0	NA	100.0%
healthy ovulatory	36-month full diary	Herzog C1/C2 plus >=4 seizure-day minimum	48,359	4,940	10.2%	3.3%
healthy ovulatory	36-month full diary	Negative-binomial regression C1/C2	48,359	1,966	4.1%	3.3%
healthy ovulatory	36-month full diary	Windowed Herzog C1/C2 union	49,605	5,576	11.2%	0.8%
healthy ovulatory	36-month full diary	Windowed Herzog thresholds	49,605	5,576	11.2%	0.8%
heterogeneous menstruating-age	1 month	Cycle reproducibility >=2/3, C1/C2	0	0	NA	100.0%
heterogeneous menstruating-age	1 month	Exact Herzog 2004, any pattern	0	0	NA	100.0%

Cohort	Observation window	Definition	Classifiable windows	False-positive windows	False-positive rate	Indeterminate
heterogeneous menstruating-age	1 month	Herzog C1/C2 plus >=4 seizure-day minimum	0	0	NA	100.0%
heterogeneous menstruating-age	1 month	Negative-binomial regression C1/C2	0	0	NA	100.0%
heterogeneous menstruating-age	1 month	Windowed Herzog C1/C2 union	38,083	18,300	48.1%	23.8%
heterogeneous menstruating-age	1 month	Windowed Herzog thresholds	38,083	19,748	51.9%	23.8%
heterogeneous menstruating-age	3 months	Cycle reproducibility >=2/3, C1/C2	0	0	NA	100.0%
heterogeneous menstruating-age	3 months	Exact Herzog 2004, any pattern	0	0	NA	100.0%
heterogeneous menstruating-age	3 months	Herzog C1/C2 plus >=4 seizure-day minimum	0	0	NA	100.0%
heterogeneous menstruating-age	3 months	Negative-binomial regression C1/C2	0	0	NA	100.0%
heterogeneous menstruating-age	3 months	Windowed Herzog C1/C2 union	45,297	19,056	42.1%	9.4%
heterogeneous menstruating-age	3 months	Windowed Herzog thresholds	45,297	23,184	51.2%	9.4%
heterogeneous menstruating-age	4 months	Cycle reproducibility >=2/3, C1/C2	0	0	NA	100.0%
heterogeneous menstruating-age	4 months	Exact Herzog 2004, any pattern	0	0	NA	100.0%
heterogeneous menstruating-age	4 months	Herzog C1/C2 plus >=4 seizure-day minimum	37,530	13,732	36.6%	24.9%
heterogeneous menstruating-age	4 months	Negative-binomial regression C1/C2	0	0	NA	100.0%
heterogeneous menstruating-age	4 months	Windowed Herzog C1/C2 union	46,369	18,279	39.4%	7.3%
heterogeneous menstruating-age	4 months	Windowed Herzog thresholds	46,369	23,060	49.7%	7.3%

Cohort	Observation window	Definition	Classifiable windows	False-positive windows	False-positive rate	Indeterminate
heterogeneous menstruating-age	6 months	Cycle reproducibility >=2/3, C1/C2	5,189	401	7.7%	89.6%
heterogeneous menstruating-age	6 months	Exact Herzog 2004, any pattern	0	0	NA	100.0%
heterogeneous menstruating-age	6 months	Herzog C1/C2 plus >=4 seizure-day minimum	41,240	13,284	32.2%	17.5%
heterogeneous menstruating-age	6 months	Negative-binomial regression C1/C2	0	0	NA	100.0%
heterogeneous menstruating-age	6 months	Windowed Herzog C1/C2 union	47,490	16,567	34.9%	5.0%
heterogeneous menstruating-age	6 months	Windowed Herzog thresholds	47,490	22,546	47.5%	5.0%
heterogeneous menstruating-age	9 months	Cycle reproducibility >=2/3, C1/C2	21,809	378	1.7%	56.4%
heterogeneous menstruating-age	9 months	Exact Herzog 2004, any pattern	0	0	NA	100.0%
heterogeneous menstruating-age	9 months	Herzog C1/C2 plus >=4 seizure-day minimum	43,843	11,844	27.0%	12.3%
heterogeneous menstruating-age	9 months	Negative-binomial regression C1/C2	0	0	NA	100.0%
heterogeneous menstruating-age	9 months	Windowed Herzog C1/C2 union	48,315	14,234	29.5%	3.4%
heterogeneous menstruating-age	9 months	Windowed Herzog thresholds	48,315	21,444	44.4%	3.4%
heterogeneous menstruating-age	12 months	Cycle reproducibility >=2/3, C1/C2	19,802	138	0.7%	60.4%
heterogeneous menstruating-age	12 months	Exact Herzog 2004, any pattern	0	0	NA	100.0%
heterogeneous menstruating-age	12 months	Herzog C1/C2 plus >=4 seizure-day minimum	45,331	10,921	24.1%	9.3%
heterogeneous menstruating-age	12 months	Negative-binomial regression C1/C2	0	0	NA	100.0%

Cohort	Observation window	Definition	Classifiable windows	False-positive windows	False-positive rate	Indeterminate
heterogeneous menstruating-age	12 months	Windowed Herzog C1/C2 union	48,713	12,713	26.1%	2.6%
heterogeneous menstruating-age	12 months	Windowed Herzog thresholds	48,713	20,738	42.6%	2.6%
heterogeneous menstruating-age	18 months	Cycle reproducibility >=2/3, C1/C2	15,646	24	0.2%	68.7%
heterogeneous menstruating-age	18 months	Exact Herzog 2004, any pattern	0	0	NA	100.0%
heterogeneous menstruating-age	18 months	Herzog C1/C2 plus >=4 seizure-day minimum	46,739	8,710	18.6%	6.5%
heterogeneous menstruating-age	18 months	Negative-binomial regression C1/C2	0	0	NA	100.0%
heterogeneous menstruating-age	18 months	Windowed Herzog C1/C2 union	49,143	9,955	20.3%	1.7%
heterogeneous menstruating-age	18 months	Windowed Herzog thresholds	49,143	19,559	39.8%	1.7%
heterogeneous menstruating-age	24 months	Cycle reproducibility >=2/3, C1/C2	12,938	3	0.0%	74.1%
heterogeneous menstruating-age	24 months	Exact Herzog 2004, any pattern	0	0	NA	100.0%
heterogeneous menstruating-age	24 months	Herzog C1/C2 plus >=4 seizure-day minimum	47,522	7,151	15.0%	5.0%
heterogeneous menstruating-age	24 months	Negative-binomial regression C1/C2	0	0	NA	100.0%
heterogeneous menstruating-age	24 months	Windowed Herzog C1/C2 union	49,343	8,112	16.4%	1.3%
heterogeneous menstruating-age	24 months	Windowed Herzog thresholds	49,343	18,615	37.7%	1.3%
heterogeneous menstruating-age	36 months	Cycle reproducibility >=2/3, C1/C2	9,243	0	0.0%	81.5%
heterogeneous menstruating-age	36 months	Exact Herzog 2004, any pattern	0	0	NA	100.0%

Cohort	Observation window	Definition	Classifiable windows	False-positive windows	False-positive rate	Indeterminate
heterogeneous menstruating-age	36 months	Herzog C1/C2 plus >=4 seizure-day minimum	48,333	5,183	10.7%	3.3%
heterogeneous menstruating-age	36 months	Negative-binomial regression C1/C2	0	0	NA	100.0%
heterogeneous menstruating-age	36 months	Windowed Herzog C1/C2 union	49,587	5,837	11.8%	0.8%
heterogeneous menstruating-age	36 months	Windowed Herzog thresholds	49,587	17,983	36.3%	0.8%
heterogeneous menstruating-age	3 cycles	Cycle reproducibility >=2/3, C1/C2	0	0	NA	100.0%
heterogeneous menstruating-age	3 cycles	Exact Herzog 2004, any pattern	16,757	8,576	51.2%	66.5%
heterogeneous menstruating-age	3 cycles	Herzog C1/C2 plus >=4 seizure-day minimum	2,015	744	36.9%	96.0%
heterogeneous menstruating-age	3 cycles	Negative-binomial regression C1/C2	0	0	NA	100.0%
heterogeneous menstruating-age	3 cycles	Windowed Herzog C1/C2 union	45,266	18,451	40.8%	9.5%
heterogeneous menstruating-age	3 cycles	Windowed Herzog thresholds	45,266	22,640	50.0%	9.5%
heterogeneous menstruating-age	6 cycles	Cycle reproducibility >=2/3, C1/C2	25,545	1,639	6.4%	48.9%
heterogeneous menstruating-age	6 cycles	Exact Herzog 2004, any pattern	0	0	NA	100.0%
heterogeneous menstruating-age	6 cycles	Herzog C1/C2 plus >=4 seizure-day minimum	41,264	12,885	31.2%	17.5%
heterogeneous menstruating-age	6 cycles	Negative-binomial regression C1/C2	0	0	NA	100.0%
heterogeneous menstruating-age	6 cycles	Windowed Herzog C1/C2 union	47,560	16,143	33.9%	4.9%
heterogeneous menstruating-age	6 cycles	Windowed Herzog thresholds	47,560	22,129	46.5%	4.9%

Cohort	Observation window	Definition	Classifiable windows	False-positive windows	False-positive rate	Indeterminate
heterogeneous menstruating-age	12 cycles	Cycle reproducibility >=2/3, C1/C2	18,764	168	0.9%	62.5%
heterogeneous menstruating-age	12 cycles	Exact Herzog 2004, any pattern	0	0	NA	100.0%
heterogeneous menstruating-age	12 cycles	Herzog C1/C2 plus >=4 seizure-day minimum	45,354	10,560	23.3%	9.3%
heterogeneous menstruating-age	12 cycles	Negative-binomial regression C1/C2	0	0	NA	100.0%
heterogeneous menstruating-age	12 cycles	Windowed Herzog C1/C2 union	48,699	12,291	25.2%	2.6%
heterogeneous menstruating-age	12 cycles	Windowed Herzog thresholds	48,699	20,319	41.7%	2.6%
heterogeneous menstruating-age	36-month full diary	Cycle reproducibility >=2/3, C1/C2	9,243	0	0.0%	81.5%
heterogeneous menstruating-age	36-month full diary	Exact Herzog 2004, any pattern	0	0	NA	100.0%
heterogeneous menstruating-age	36-month full diary	Herzog C1/C2 plus >=4 seizure-day minimum	48,333	5,183	10.7%	3.3%
heterogeneous menstruating-age	36-month full diary	Negative-binomial regression C1/C2	48,333	2,032	4.2%	3.3%
heterogeneous menstruating-age	36-month full diary	Windowed Herzog C1/C2 union	49,587	5,837	11.8%	0.8%
heterogeneous menstruating-age	36-month full diary	Windowed Herzog thresholds	49,587	17,983	36.3%	0.8%

Rows with fewer than 1,000 classifiable windows are retained for transparency but their positive counts and rates are suppressed to avoid over-interpretation.

Supplementary Table 4. Phase-labeling sensitivity.

Cohort	Phase mode	Observation window	Definition	Classifiable / attempted	Positive among classifiable	Positive among all attempted	Indeterminate
healthy ovulatory	Luteal-anchored ovulatory	3 months	Windowed Herzog C1/C2 union	45,280 / 50,000	39.6% (39.1%-40.0%)	35.9%	9.4%
healthy ovulatory	Luteal-anchored ovulatory	3 months	Windowed Herzog thresholds	45,280 / 50,000	39.6% (39.1%-40.0%)	35.9%	9.4%
healthy ovulatory	Luteal-anchored ovulatory	12 months	Windowed Herzog C1/C2 union	48,754 / 50,000	26.0% (25.6%-26.4%)	25.4%	2.5%
healthy ovulatory	Luteal-anchored ovulatory	12 months	Windowed Herzog thresholds	48,754 / 50,000	26.0% (25.6%-26.4%)	25.4%	2.5%
healthy ovulatory	Luteal-anchored ovulatory	36-month full diary	Windowed Herzog C1/C2 union	49,605 / 50,000	12.2% (11.9%-12.5%)	12.1%	0.8%
healthy ovulatory	Luteal-anchored ovulatory	36-month full diary	Windowed Herzog thresholds	49,605 / 50,000	12.2% (11.9%-12.5%)	12.1%	0.8%
healthy ovulatory	Strict Herzog	3 months	Windowed Herzog C1/C2 union	45,280 / 50,000	41.3% (40.8%-41.7%)	37.4%	9.4%
healthy ovulatory	Strict Herzog	3 months	Windowed Herzog thresholds	45,280 / 50,000	41.3% (40.8%-41.7%)	37.4%	9.4%
healthy ovulatory	Strict Herzog	12 months	Windowed Herzog C1/C2 union	48,754 / 50,000	25.6% (25.2%-26.0%)	25.0%	2.5%
healthy ovulatory	Strict Herzog	12 months	Windowed Herzog thresholds	48,754 / 50,000	25.6% (25.2%-26.0%)	25.0%	2.5%
healthy ovulatory	Strict Herzog	36-month full diary	Windowed Herzog C1/C2 union	49,605 / 50,000	11.2% (11.0%-11.5%)	11.2%	0.8%
healthy ovulatory	Strict Herzog	36-month full diary	Windowed Herzog thresholds	49,605 / 50,000	11.2% (11.0%-11.5%)	11.2%	0.8%
heterogeneous menstruating-age	Luteal-anchored ovulatory	3 months	Windowed Herzog C1/C2 union	45,297 / 50,000	40.3% (39.9%-40.8%)	36.5%	9.4%
heterogeneous menstruating-age	Luteal-anchored ovulatory	3 months	Windowed Herzog thresholds	45,297 / 50,000	46.9% (46.4%-47.4%)	42.5%	9.4%
heterogeneous menstruating-age	Luteal-anchored ovulatory	12 months	Windowed Herzog C1/C2 union	48,713 / 50,000	26.9% (26.5%-27.3%)	26.2%	2.6%
heterogeneous menstruating-age	Luteal-anchored ovulatory	12 months	Windowed Herzog thresholds	48,713 / 50,000	38.1% (37.7%-38.6%)	37.1%	2.6%

Cohort	Phase mode	Observation window	Definition	Classifiable / attempted	Positive among classifiable	Positive among all attempted	Indeterminate
heterogeneous menstruating-age	Luteal-anchored ovulatory	36-month full diary	Windowed Herzog C1/C2 union	49,587 / 50,000	12.8% (12.5%-13.1%)	12.7%	0.8%
heterogeneous menstruating-age	Luteal-anchored ovulatory	36-month full diary	Windowed Herzog thresholds	49,587 / 50,000	30.0% (29.6%-30.4%)	29.8%	0.8%
heterogeneous menstruating-age	Strict Herzog	3 months	Windowed Herzog C1/C2 union	45,297 / 50,000	42.1% (41.6%-42.5%)	38.1%	9.4%
heterogeneous menstruating-age	Strict Herzog	3 months	Windowed Herzog thresholds	45,297 / 50,000	51.2% (50.7%-51.6%)	46.4%	9.4%
heterogeneous menstruating-age	Strict Herzog	12 months	Windowed Herzog C1/C2 union	48,713 / 50,000	26.1% (25.7%-26.5%)	25.4%	2.6%
heterogeneous menstruating-age	Strict Herzog	12 months	Windowed Herzog thresholds	48,713 / 50,000	42.6% (42.1%-43.0%)	41.5%	2.6%
heterogeneous menstruating-age	Strict Herzog	36-month full diary	Windowed Herzog C1/C2 union	49,587 / 50,000	11.8% (11.5%-12.1%)	11.7%	0.8%
heterogeneous menstruating-age	Strict Herzog	36-month full diary	Windowed Herzog thresholds	49,587 / 50,000	36.3% (35.8%-36.7%)	36.0%	0.8%

Supplementary Table 5. C1/C2/C3 decomposition and C3-exclusion sensitivity.

Cohort	Definition	Classifiable windows	Positive windows	Apparent classification rate	Indeterminate
healthy ovulatory	Herzog plus >=4 seizure-day minimum	48,359	4,940	10.2% (9.9%-10.5%)	3.3%
healthy ovulatory	Herzog plus >=4 seizure-day minimum, excluding C3	48,359	4,940	10.2% (9.9%-10.5%)	3.3%
healthy ovulatory	Windowed Herzog C1 only	49,451	3,893	7.9% (7.6%-8.1%)	1.1%
healthy ovulatory	Windowed Herzog C2 only	49,495	2,753	5.6% (5.4%-5.8%)	1.0%
healthy ovulatory	Windowed Herzog C3 only	0	0	NA (NA-NA)	100.0%
healthy ovulatory	Windowed Herzog thresholds	49,605	5,576	11.2% (11.0%-11.5%)	0.8%
healthy ovulatory	Windowed Herzog thresholds, excluding C3	49,605	5,576	11.2% (11.0%-11.5%)	0.8%

Cohort	Definition	Classifiable windows	Positive windows	Apparent classification rate	Indeterminate
heterogeneous menstruating-age	Herzog plus >=4 seizure-day minimum	48,333	17,206	35.6% (35.2%-36.0%)	3.3%
heterogeneous menstruating-age	Herzog plus >=4 seizure-day minimum, excluding C3	48,333	5,183	10.7% (10.5%-11.0%)	3.3%
heterogeneous menstruating-age	Windowed Herzog C1 only	49,392	4,171	8.4% (8.2%-8.7%)	1.2%
heterogeneous menstruating-age	Windowed Herzog C2 only	49,482	2,788	5.6% (5.4%-5.8%)	1.0%
heterogeneous menstruating-age	Windowed Herzog C3 only	38,989	14,425	37.0% (36.5%-37.5%)	22.0%
heterogeneous menstruating-age	Windowed Herzog thresholds	49,587	17,983	36.3% (35.8%-36.7%)	0.8%
heterogeneous menstruating-age	Windowed Herzog thresholds, excluding C3	49,587	5,837	11.8% (11.5%-12.1%)	0.8%

Supplementary Table 6. Negative-binomial stabilized versus window-only dispersion sensitivity.

Cohort	Observation window	Regression comparator	Classifiable windows	Apparent classification rate	Indeterminate
healthy ovulatory	36-month full diary	Negative-binomial regression C1/C2	48,359	4.1% (3.9%-4.2%)	3.3%
healthy ovulatory	36-month full diary	Negative-binomial regression C1/C2, window-only dispersion	48,359	4.1% (3.9%-4.2%)	3.3%
heterogeneous menstruating-age	36-month full diary	Negative-binomial regression C1/C2	48,333	4.2% (4.0%-4.4%)	3.3%
heterogeneous menstruating-age	36-month full diary	Negative-binomial regression C1/C2, window-only dispersion	48,333	4.2% (4.0%-4.4%)	3.3%

Supplementary Table 7. Study-level Monte Carlo results.

Cohort	N per study	Definition	Denominator	Mean apparent prevalence	2.5th-97.5th percentiles	P >=39.1%	P >=44.2%
healthy ovulatory	30	Windowed Herzog C1/C2 union	All participants	37.5%	20.0%-53.3%	45.4%	20.1%
healthy ovulatory	30	Windowed Herzog C1/C2 union	Classifiable participants only	41.4%	23.1%-60.0%	59.9%	38.4%
healthy ovulatory	30	Windowed Herzog thresholds	All participants	37.5%	20.0%-53.3%	45.4%	20.1%
healthy ovulatory	30	Windowed Herzog thresholds	Classifiable participants only	41.4%	23.1%-60.0%	59.9%	38.4%
healthy ovulatory	50	Windowed Herzog C1/C2 union	All participants	37.3%	24.0%-50.0%	39.7%	13.0%
healthy ovulatory	50	Windowed Herzog C1/C2 union	Classifiable participants only	41.3%	27.3%-56.1%	62.4%	34.3%
healthy ovulatory	50	Windowed Herzog thresholds	All participants	37.3%	24.0%-50.0%	39.7%	13.0%
healthy ovulatory	50	Windowed Herzog thresholds	Classifiable participants only	41.3%	27.3%-56.1%	62.4%	34.3%
healthy ovulatory	100	Windowed Herzog C1/C2 union	All participants	37.5%	28.0%-47.0%	33.8%	7.1%
healthy ovulatory	100	Windowed Herzog C1/C2 union	Classifiable participants only	41.4%	31.5%-51.6%	67.0%	29.1%
healthy ovulatory	100	Windowed Herzog thresholds	All participants	37.5%	28.0%-47.0%	33.8%	7.1%
healthy ovulatory	100	Windowed Herzog thresholds	Classifiable participants only	41.4%	31.5%-51.6%	67.0%	29.1%
heterogeneous menstruating-age	30	Windowed Herzog C1/C2 union	All participants	38.2%	20.0%-56.7%	48.9%	22.4%
heterogeneous menstruating-age	30	Windowed Herzog C1/C2 union	Classifiable participants only	42.2%	24.0%-60.7%	63.5%	42.3%
heterogeneous menstruating-age	30	Windowed Herzog thresholds	All participants	46.3%	30.0%-63.3%	80.5%	55.9%
heterogeneous menstruating-age	30	Windowed Herzog thresholds	Classifiable participants only	51.0%	32.1%-69.2%	89.7%	76.6%

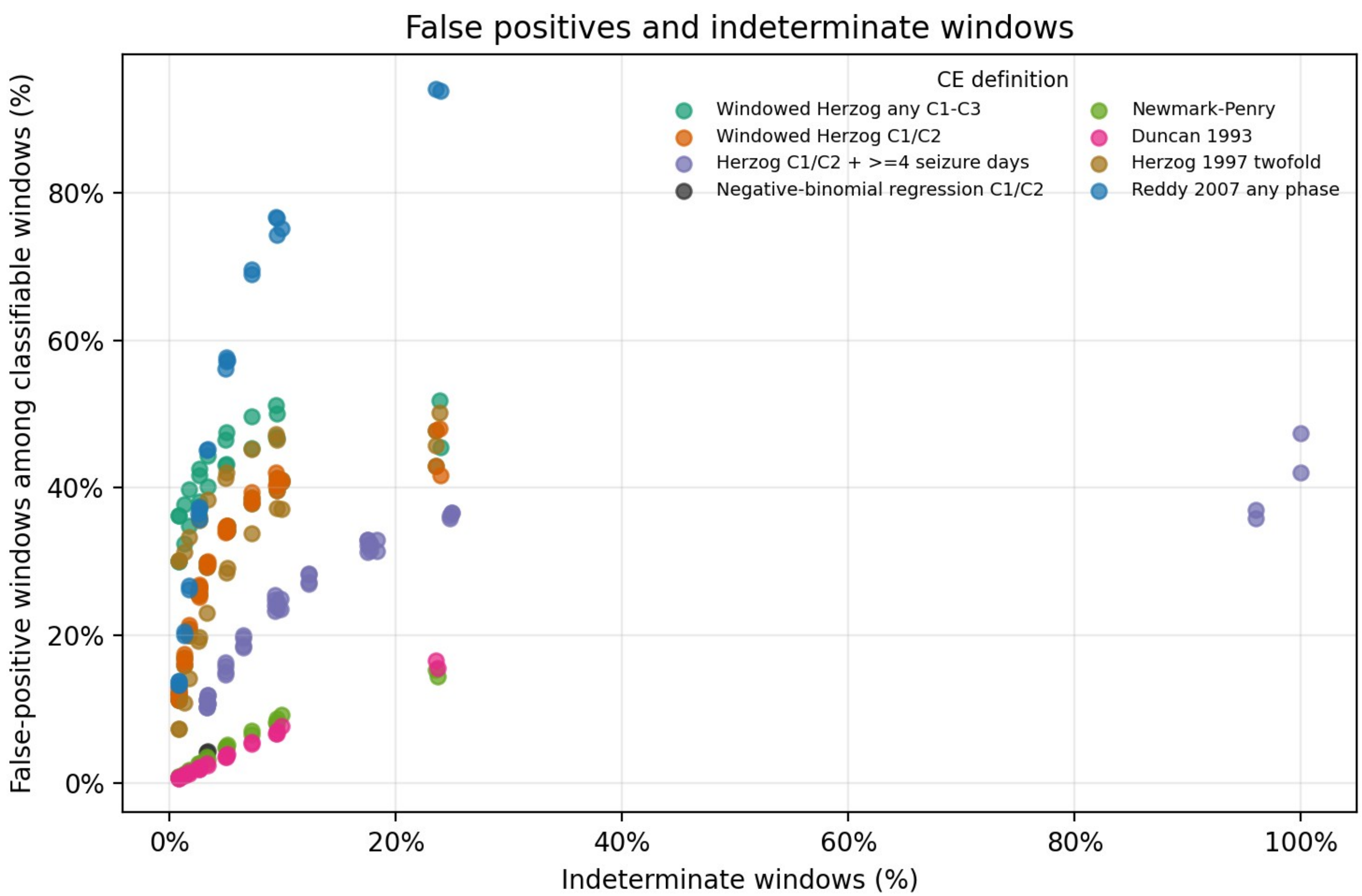
Cohort	N per study	Definition	Denominator	Mean apparent prevalence	2.5th-97.5th percentiles	P >=39.1%	P >=44.2%
heterogeneous menstruating-age	50	Windowed Herzog C1/C2 union	All participants	38.1%	26.0%-52.0%	43.9%	15.3%
heterogeneous menstruating-age	50	Windowed Herzog C1/C2 union	Classifiable participants only	42.0%	28.3%-56.5%	65.6%	38.0%
heterogeneous menstruating-age	50	Windowed Herzog thresholds	All participants	46.1%	32.0%-60.0%	84.5%	55.9%
heterogeneous menstruating-age	50	Windowed Herzog thresholds	Classifiable participants only	50.9%	36.2%-65.9%	94.9%	81.4%
heterogeneous menstruating-age	100	Windowed Herzog C1/C2 union	All participants	38.0%	29.0%-47.0%	38.2%	8.8%
heterogeneous menstruating-age	100	Windowed Herzog C1/C2 union	Classifiable participants only	42.0%	31.9%-52.1%	71.5%	33.1%
heterogeneous menstruating-age	100	Windowed Herzog thresholds	All participants	46.0%	36.0%-56.0%	90.8%	62.2%
heterogeneous menstruating-age	100	Windowed Herzog thresholds	Classifiable participants only	50.9%	40.4%-61.1%	98.7%	89.7%

Supplementary Table 8. Assumption-based historical definitions.

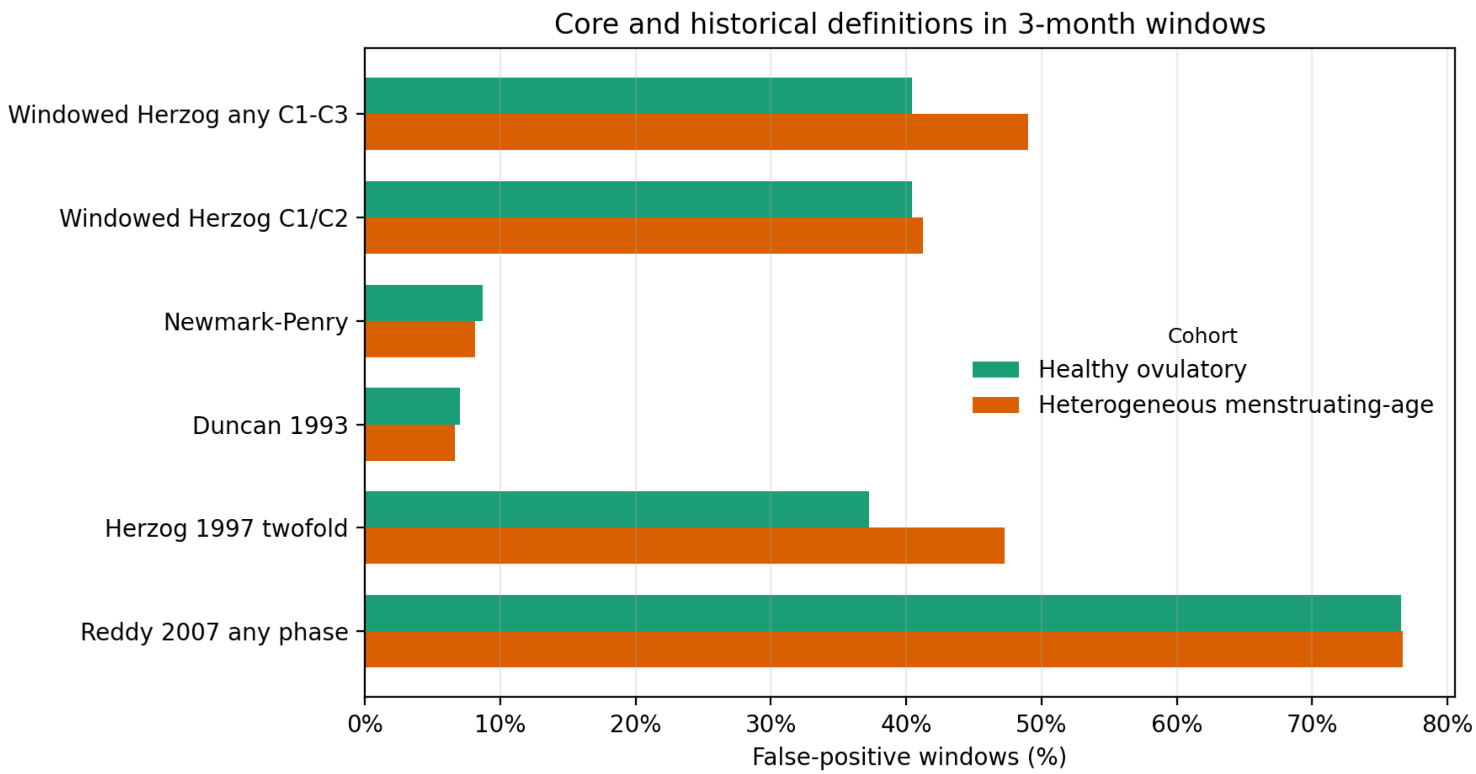
Cohort	Observation window	Definition	Classifiable windows	False-positive windows	False-positive rate	Indeterminate
healthy ovulatory	3 months	Duncan 1993 ten-day rule	45,280	3,163	7.0% (6.8%-7.2%)	9.4%
healthy ovulatory	3 months	Herzog 1997 twofold rule	45,280	16,862	37.2% (36.8%-37.7%)	9.4%
healthy ovulatory	3 months	Newmark-Penry perimenstrual rule	45,280	3,939	8.7% (8.4%-9.0%)	9.4%
healthy ovulatory	3 months	Newmark-Penry two-thirds sensitivity	45,280	1,870	4.1% (4.0%-4.3%)	9.4%
healthy ovulatory	3 months	Reddy 2007 any-phase twofold rule	45,280	34,681	76.6% (76.2%-77.0%)	9.4%
healthy ovulatory	36-month full diary	Duncan 1993 ten-day rule	49,605	361	0.7% (0.7%-0.8%)	0.8%
healthy ovulatory	36-month full diary	Herzog 1997 twofold rule	49,605	3,603	7.3% (7.0%-7.5%)	0.8%

Cohort	Observation window	Definition	Classifiable windows	False-positive windows	False-positive rate	Indeterminate
healthy ovulatory	36-month full diary	Newmark-Penry perimenstrual rule	49,605	427	0.9% (0.8%-0.9%)	0.8%
healthy ovulatory	36-month full diary	Newmark-Penry two-thirds sensitivity	49,605	170	0.3% (0.3%-0.4%)	0.8%
healthy ovulatory	36-month full diary	Reddy 2007 any-phase twofold rule	49,605	6,599	13.3% (13.0%-13.6%)	0.8%
heterogeneous menstruating-age	3 months	Duncan 1993 ten-day rule	45,297	3,018	6.7% (6.4%-6.9%)	9.4%
heterogeneous menstruating-age	3 months	Herzog 1997 twofold rule	45,297	21,417	47.3% (46.8%-47.7%)	9.4%
heterogeneous menstruating-age	3 months	Newmark-Penry perimenstrual rule	45,297	3,696	8.2% (7.9%-8.4%)	9.4%
heterogeneous menstruating-age	3 months	Newmark-Penry two-thirds sensitivity	45,297	1,733	3.8% (3.7%-4.0%)	9.4%
heterogeneous menstruating-age	3 months	Reddy 2007 any-phase twofold rule	45,297	34,750	76.7% (76.3%-77.1%)	9.4%
heterogeneous menstruating-age	36-month full diary	Duncan 1993 ten-day rule	49,587	311	0.6% (0.6%-0.7%)	0.8%
heterogeneous menstruating-age	36-month full diary	Herzog 1997 twofold rule	49,587	14,914	30.1% (29.7%-30.5%)	0.8%
heterogeneous menstruating-age	36-month full diary	Newmark-Penry perimenstrual rule	49,587	378	0.8% (0.7%-0.8%)	0.8%
heterogeneous menstruating-age	36-month full diary	Newmark-Penry two-thirds sensitivity	49,587	175	0.4% (0.3%-0.4%)	0.8%
heterogeneous menstruating-age	36-month full diary	Reddy 2007 any-phase twofold rule	49,587	6,854	13.8% (13.5%-14.1%)	0.8%

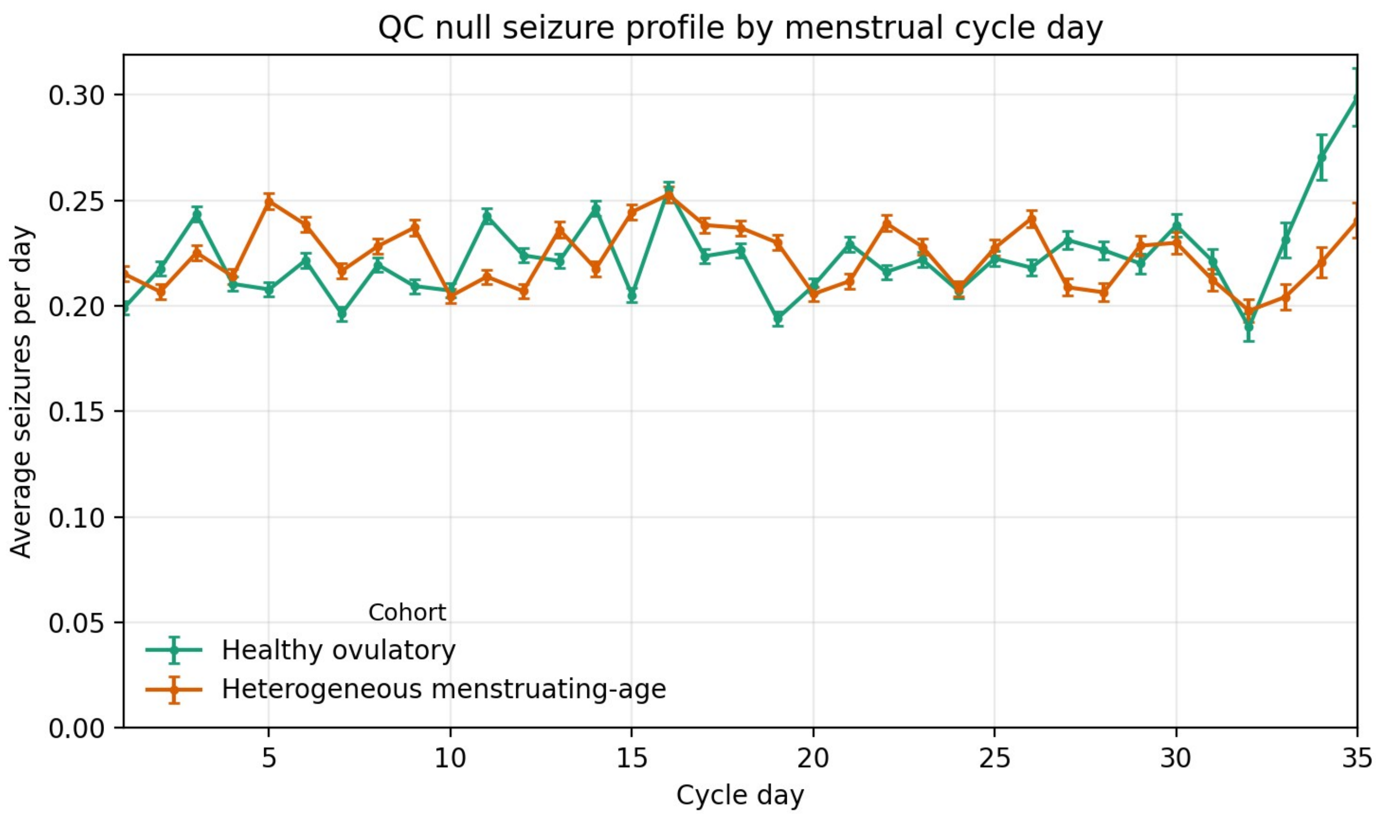
Supplementary Figure 1. False-positive and indeterminate frontier.



Supplementary Figure 2. Historical and core definitions in 3-month windows.



Supplementary Figure 3. Null seizure profile by menstrual cycle day.



A8. Output provenance and assumptions

- Definition D uses a participant-full-diary method-of-moments negative-binomial alpha recorded in d_alpha; Poisson robust fallback is recorded in d_reason when statsmodels negative-binomial fitting fails. D_window_alpha re-estimates alpha from the analyzed window as a non-oracle sensitivity.

- Healthy ovulatory cohort used hormone_cycler build_patient_profile/render_cycle with ovulation_probability set to 1.0 because simulate_diary did not expose a public force-ovulation knob.
- Historical definitions H1-H4 are assumption-based operationalizations and are flagged in summary outputs.
- Study-level Monte Carlo samples each selected participant from a deterministic pool of precomputed random valid 3-month windows to avoid retaining all daily diaries in memory.
- The hormone simulator exposes medical-factor controls but no natural prevalence sampler; heterogeneous-cohort medical factors were sampled from config.yaml rates.
- True-coupling code paths are retained for future operating-characteristic analyses but are not reported as part of this null-only Paper 1 revision.